

Database Queries-02-52

We have change the database table name employee to emp

The screenshot shows the SQL Developer interface. The query window contains the following SQL commands:

```
88 • select Dept, avg(Salary) as AvgSal from (Select Dept, Salary from Salary where Deptid = 201) as Salary group by Dept;
89
90 • show databases;
91 • use employee;
92 • rename table employee to emp;
93 • select * from emp;
```

Red arrows point to the 'rename table employee to emp;' and 'select * from emp;' lines in the query window.

The result grid displays the following data:

Employeeid	Firstname	Lastname	Age	Gender	Birthdate	Phonenumber	Location	Maritalstatus
2345	Nushima	Noushadh	21	Female	2004-05-01	987654321	Coimbatore	Single
2346	Tannishtha	Saravana	23	Female	2002-01-05	98787643	Ooty	Single
2347	Vayakal	Pladhavan	25	Male	2000-10-10	123456789	Chitoor	Married
2348	Sangam	Srivelli	28	Female	2004-11-02	678954321	Kakkonada	Single
2349	Sheshadri	Nagendran	30	Male	1995-01-01	98765776	Chitoor	Single
2350	Bharath	Mummadi	20	Male	2004-12-01	996632265	hyderabad	Single

Red arrows point to the 'Result Grid' button and the 'Apply' button.

The output window shows the following messages:

#	Time	Action	Message
32	12:38:47	use employee	0 row(s) affected
33	12:48:10	rename table employee to emp	0 row(s) affected
34	12:48:17	show databases	7 row(s) returned
35	12:48:31	use employee	0 row(s) affected
36	12:49:00	select * from employee LIMIT 0, 1000	Error Code: 1146. Table 'employee.employee' doesn't exist
37	12:49:07	select * from emp LIMIT 0, 1000	11 row(s) returned

Red arrows point to the 'rename table employee to emp' and 'select * from emp' messages in the output window.

We have truncated everything from emp table and selected it to see the data is their or not

Query 1

```

89
90 • show databases;
91 • use employee;
92 • rename table employee to emp;
93 • select * from emp;
94 • truncate table emp;
95
96

```

Result Grid

Employeeid	Firstname	Lastname	Age	Gender	Birthdate	Phonenumber	Location	Maritalstatus
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Output

#	Time	Action	Message
34	12:48:17	show databases	7 row(s) returned
35	12:48:31	use employee	0 row(s) affected
36	12:49:00	select * from employee LIMIT 0, 1000	Error Code: 1146. Table 'employee.employee' doesn't exist
37	12:49:07	select * from emp LIMIT 0, 1000	11 row(s) returned
38	12:52:04	truncate table emp	0 row(s) affected
39	12:52:07	select * from emp LIMIT 0, 1000	0 row(s) returned

Here we have updated a row phonenumber where employeeid

Query 1

```

89
90 • show databases;
91 • use employee;
92 • rename table employee to emp;
93 • select * from emp;
94 • truncate table emp;
95 • update emp set phonenumber=909090909 where employeeid=2355;
96

```

Result Grid

Employeeid	Firstname	Lastname	Age	Gender	Birthdate	Phonenumber	Location	Maritalstatus
2351	sandeep	sai	19	Male	2006-01-01	928371656	AP	Single
2352	kinnera	kakoli	19	Male	2006-01-01	974675283	nagapur	Single
2353	kunal	sherna	25	Male	2000-01-01	927465763	ALWAL	Single
2354	chandi	kulakarni	30	Male	1995-12-30	996737382	Hyderabad	Single
2355	vivek	amaranga...	25	Male	2000-05-24	909090909	AP	Single

Output

#	Time	Action	Message
47	12:58:20	select * from emp LIMIT 0, 1000	11 row(s) returned
48	14:05:58	update emp set phonenumber=909090909 where Employeeid=2355	Error Code: 1264. Out of range value for column 'Phonenumber' at row
49	14:06:03	select * from emp LIMIT 0, 1000	11 row(s) returned
50	14:06:32	update emp set phonenumber=909090909 where employeeid=2355	Error Code: 1264. Out of range value for column 'Phonenumber' at row
51	14:06:43	update emp set phonenumber=909090909 where employeeid=2355	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
52	14:06:47	select * from emp LIMIT 0, 1000	11 row(s) returned

Here we have deleted a row using its primary key employeeid

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'Server Enterprise1' tree with nodes for Management, Instance, and Performance. The central pane shows a query window with the following SQL script:

```

87 • select Firstname, Salary from salary where salary > all (select salary from salary where Deptid = 209);
88 • select Dept, avg(Salary) as AvgSal from (Select Dept, Salary from Salary where Deptid = 201) as Salary group by Dept;
89
90 • show databases;
91 • use employee;
92 • rename table employee to emp;
93 • select * from emp;
94 • truncate table emp;
95 • update emp set phonenum=909090909 where employeeid=2355;
96 • insert into emp (Employeeid, Firstname, Lastname, Age, Gender, Birthdate, Phonenum, Location, Maritalstatus)
97 • VALUES (2356, 'Nushima', 'Noushadh', 21, 'Female', '2004-05-01', 987654321, 'Coimbatore', 'Single');
98 • delete from emp where employeeid=2356;

```

The right pane shows the 'Output' window with the following execution results:

#	Time	Action	Message
51	14:06:43	update emp set phonenum=909090909 where employeeid=2355	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
52	14:06:47	select * from emp LIMIT 0, 1000	11 row(s) returned
53	14:10:15	insert into employee (Employeeid, Firstname, Lastname, Age, Gender, Birthdate, Phonenum, Loc...	Error Code: 1146. Table 'employee.employee' doesn't exist
54	14:11:34	insert into emp (Employeeid, Firstname, Lastname, Age, Gender, Birthdate, Phonenum, Location...	1 row(s) affected
55	14:11:41	delete from emp where Firstname='Nushima'	Error Code: 1175. You are using safe update mode and you tried to update a table witho...
56	14:14:09	delete from emp where employeeid=2356	1 row(s) affected

Red arrows point from the SQL script lines 96, 97, 98 to the corresponding messages in the output window.

We can use the comment query using alter

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'Server Enterprise1' tree with nodes for Management, Instance, and Performance. The central pane shows a query window with the following SQL script:

```

91 • use employee;
92 • rename table employee to emp;
93 • select * from emp;
94 • truncate table emp;
95 • update emp set phonenum=909090909 where employeeid=2355;
96 • insert into emp (Employeeid, Firstname, Lastname, Age, Gender, Birthdate, Phonenum, Location, Maritalstatus)
97 • VALUES (2356, 'Nushima', 'Noushadh', 21, 'Female', '2004-05-01', 987654321, 'Coimbatore', 'Single');
98 • delete from emp where employeeid=2356;
99 • alter table emp comment = 'this is the old employee table';
100

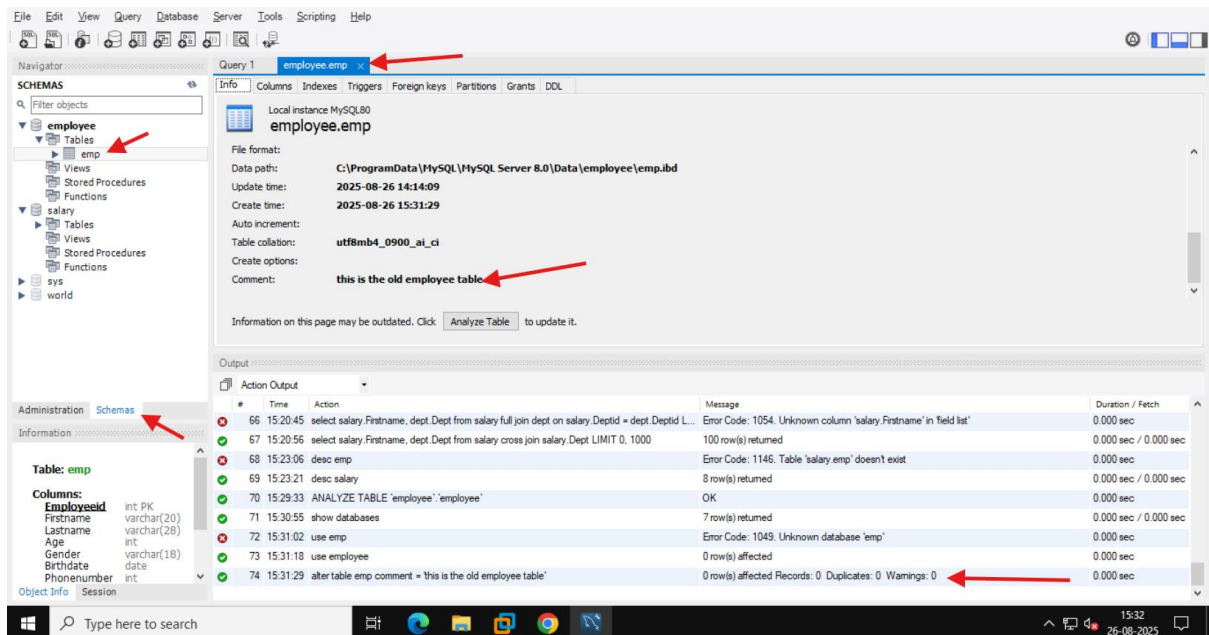
```

The right pane shows the 'Output' window with the following execution results:

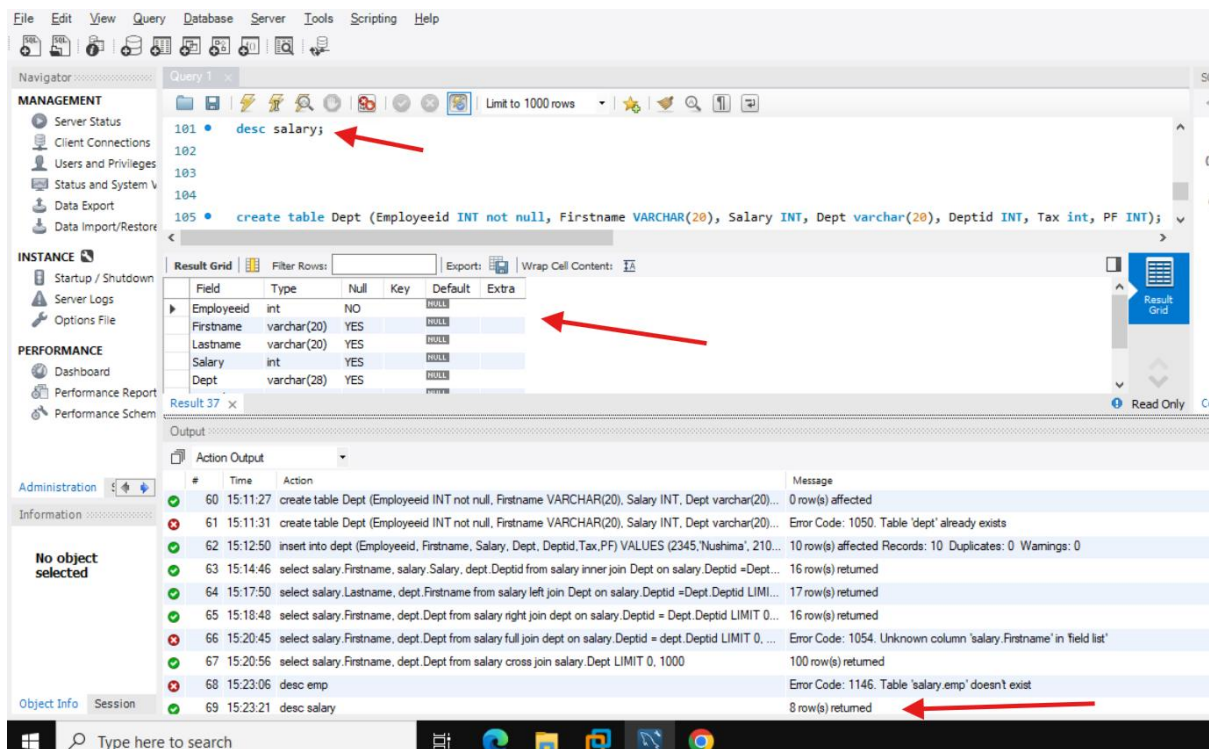
#	Time	Action	Message
46	12:58:16	use employee	0 row(s) affected
47	12:58:20	select * from emp LIMIT 0, 1000	11 row(s) returned
48	14:05:58	update emp set phonenum=909090909 where Employeeid=2355	Error Code: 1264. Out of range value for column 'Phonenum' at row 1
49	14:06:03	select * from emp LIMIT 0, 1000	11 row(s) returned
50	14:06:32	update emp set phonenum=909090909 where employeeid=2355	Error Code: 1264. Out of range value for column 'Phonenum' at row 1
51	14:06:43	update emp set phonenum=909090909 where employeeid=2355	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
52	14:06:47	select * from emp LIMIT 0, 1000	11 row(s) returned
53	14:10:15	insert into employee (Employeeid, Firstname, Lastname, Age, Gender, Birthdate, Phonenum, Loc...	Error Code: 1146. Table 'employee.employee' doesn't exist
54	14:11:34	insert into emp (Employeeid, Firstname, Lastname, Age, Gender, Birthdate, Phonenum, Location...	1 row(s) affected
55	14:11:41	delete from emp where Firstname='Nushima'	Error Code: 1175. You are using safe update mode and you tried to update a table without a
56	14:14:09	delete from emp where employeeid=2356	1 row(s) affected
57	15:09:22	alter table emp comment = 'this is the old employee table'	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

Red arrows point from the SQL script line 99 to the corresponding message in the output window.

See the comment as been added to emp table

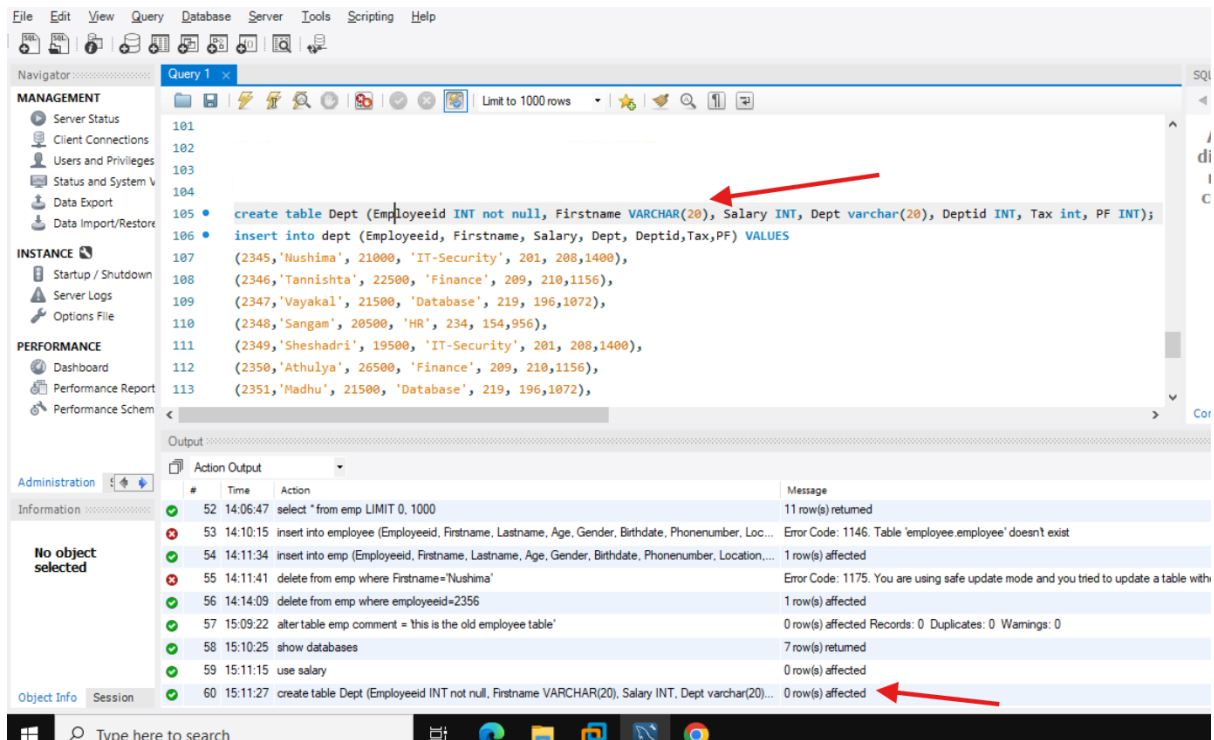


We can use Desc salary; or describe salary; to get structure or schema of the table



We are creating a new table under salary database

create table Dept (Employeeid INT not null, Firstname VARCHAR(20), Salary INT, Dept varchar(20), Deptid INT, Tax int, PF INT);



Inserting the values

insert into dept (Employeeid, Firstname, Salary, Dept, Deptid,Tax,PF) VALUES

(2345,'Nushima', 21000, 'IT-Security', 201, 208,1400),

(2346,'Tannishtha', 22500, 'Finance', 209, 210,1156),

(2347,'Vayakal', 21500, 'Database', 219, 196,1072),

(2348,'Sangam', 20500, 'HR', 234, 154,956),

(2349,'Sheshadri', 19500, 'IT-Security', 201, 208,1400),

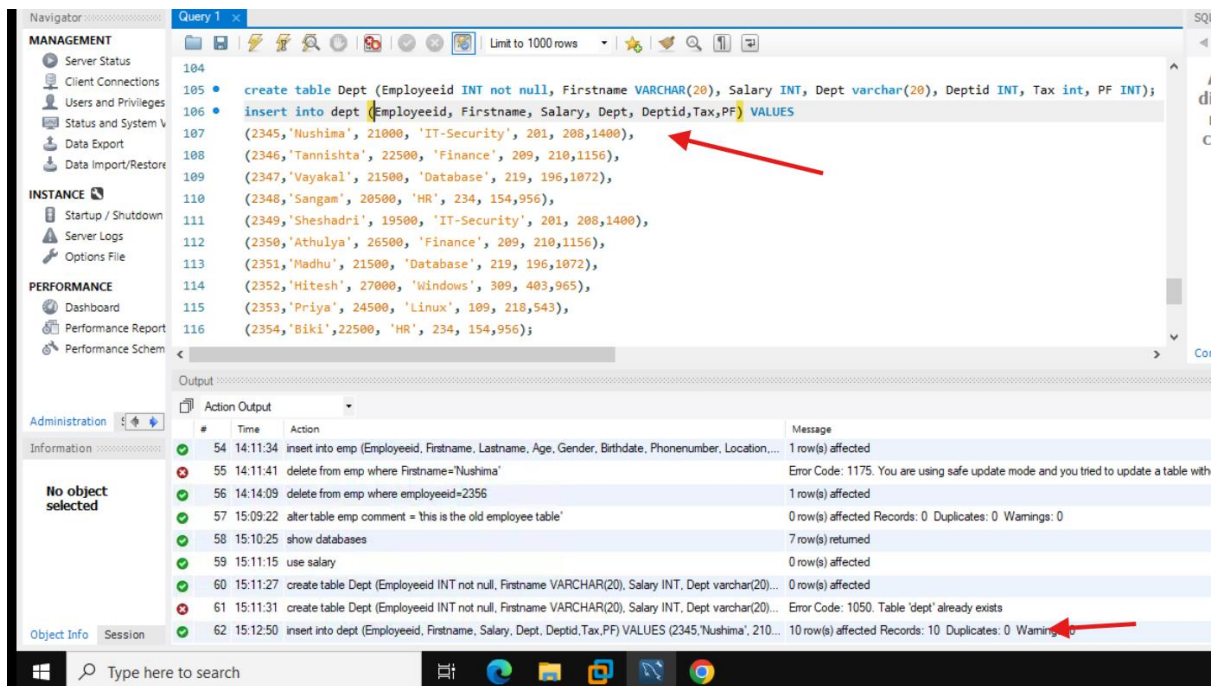
(2350,'Athulya', 26500, 'Finance', 209, 210,1156),

(2351,'Madhu', 21500, 'Database', 219, 196,1072),

(2352,'Hitesh', 27000, 'Windows', 309, 403,965),

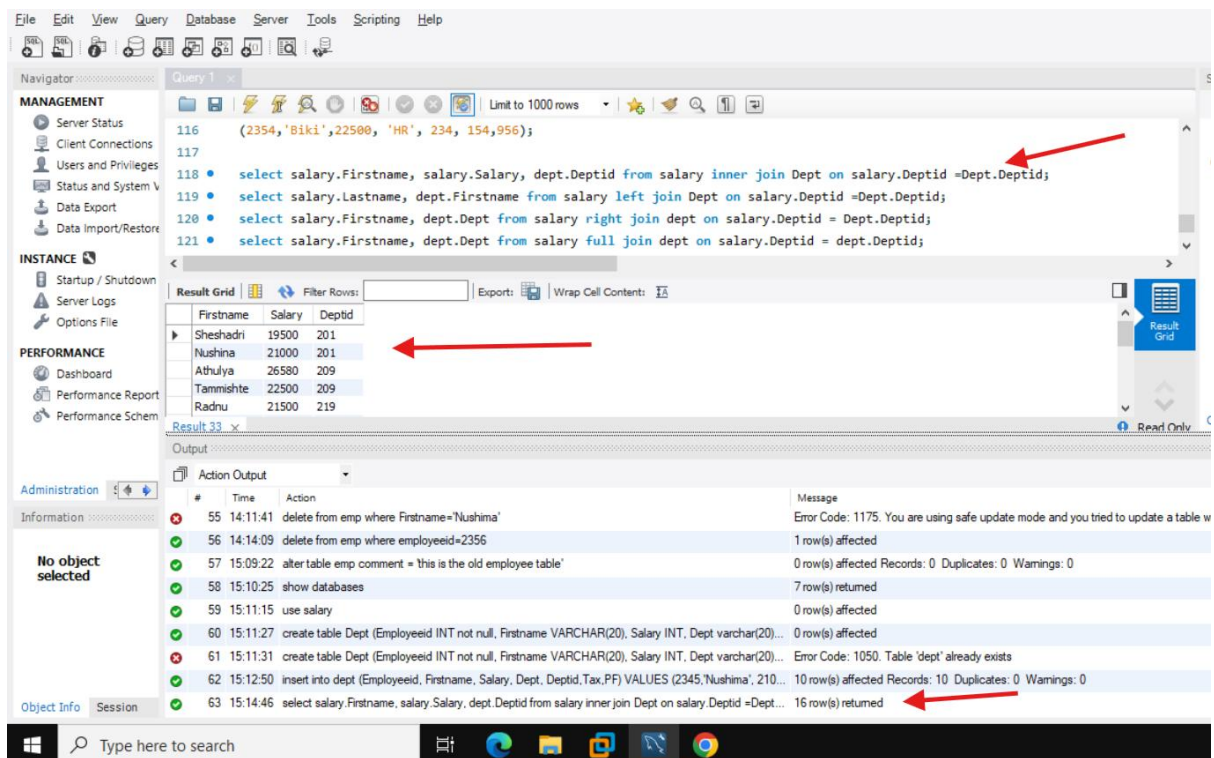
(2353,'Priya', 24500, 'Linux', 109, 218,543),

(2354,'Biki',22500, 'HR', 234, 154,956);



select salary.Firstname, salary.Salary, dept.Deptid from salary inner join Dept on salary.Deptid =Dept.Deptid;

here we are using inner join to get data from both salary and dept table



select salary.Lastname, dept.Firstname from salary left join Dept on salary.Deptid =Dept.Deptid;

Here we are using the left join to get data from left side

The screenshot shows the SQL Developer interface with a query window titled 'Query 1'. The query is as follows:

```

116 (2354,'Biki',22500,'HR', 234, 154,956);
117
118 • select salary.Firstname, salary.Salary, dept.Deptid from salary inner join Dept on salary.Deptid =Dept.Deptid;
119 • select salary.Lastname, dept.Firstname from salary left join Dept on salary.Deptid =Dept.Deptid;
120 • select salary.Firstname, dept.Dept from salary right join dept on salary.Deptid = Dept.Deptid;
121 • select salary.Firstname, dept.Dept from salary full join dept on salary.Deptid = dept.Deptid;

```

The 'Result Grid' shows the output of the right join query:

Lastname	Firstname
Noushadh	Sheshadri
Noushadh	Nushima
Saravana	Athulya
Saravana	Tannishtha
Radhavan	Madhu

The 'Action Output' window shows the execution log, with the final row indicating that the right join query returned 17 rows.

select salary.Firstname, dept.Dept from salary right join dept on salary.Deptid = Dept.Deptid;

here we are using right join

The screenshot shows the SQL Developer interface with a query window titled 'Query 1'. The query is as follows:

```

116 (2354,'Biki',22500,'HR', 234, 154,956);
117
118 • select salary.Firstname, salary.Salary, dept.Deptid from salary inner join Dept on salary.Deptid =Dept.Deptid;
119 • select salary.Lastname, dept.Firstname from salary left join Dept on salary.Deptid =Dept.Deptid;
120 • select salary.Firstname, dept.Dept from salary right join dept on salary.Deptid = Dept.Deptid;
121 • select salary.Firstname, dept.Dept from salary full join dept on salary.Deptid = dept.Deptid;

```

The 'Result Grid' shows the output of the full join query:

Firstname	Dept
Sheshadri	IT-Security
Nushima	IT-Security
Athulya	Finance
Tannishtha	Finance
Radnu	Database

The 'Action Output' window shows the execution log, with the final row indicating that the full join query returned 16 rows.

select salary.Firstname, dept.Dept from salary full join dept on salary.Deptid = dept.Deptid;

full join is not available in the mysql workbench

select salary.Firstname, dept.Dept from salary cross join salary.Dept;

The screenshot shows the MySQL Workbench interface. The 'Query' window displays a list of SQL queries. The last query, highlighted with a red arrow, is:

```
select salary.Firstname, dept.Dept from salary cross join salary.Dept;
```

The 'Result Grid' shows the output of this query, with a red arrow pointing to the first row:

Firstname	Dept
bharath	IT-Security
Priya	IT-Security
Mitesh	IT-Security
Radnu	IT-Security
Athulya	IT-Security

The 'Output' window shows the execution log. The last entry, highlighted with a red arrow, is:

#	Time	Action	Message
67	15:20:56	select salary.Firstname, dept.Dept from salary cross join salary.Dept LIMIT 0, 1000	100 row(s) returned